

AUROUAV Diagrams

AUROUAV system architecture

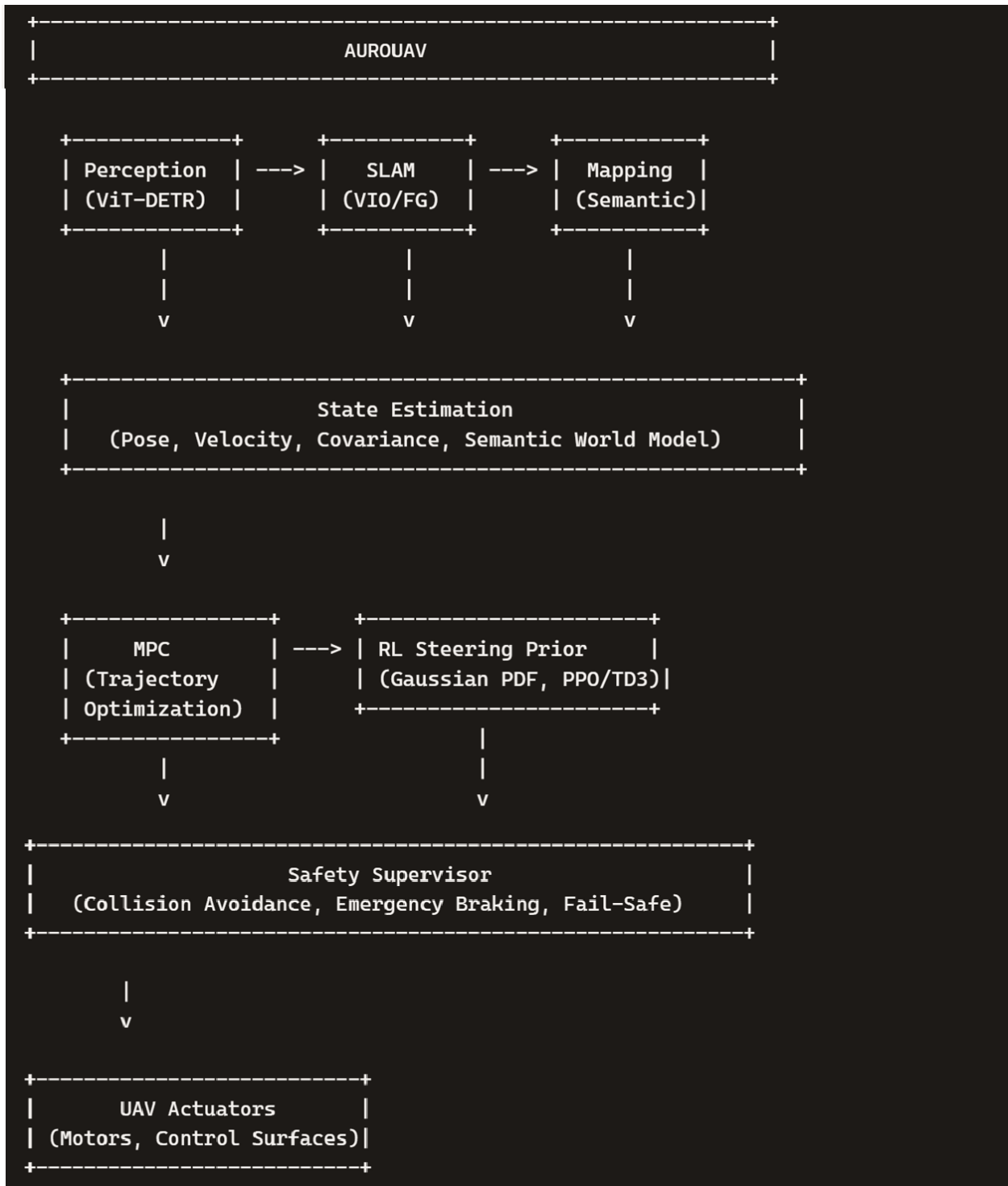


Fig. 1. AUROUAV system architecture showing the full autonomy pipeline from perception to control.

Perception pipeline

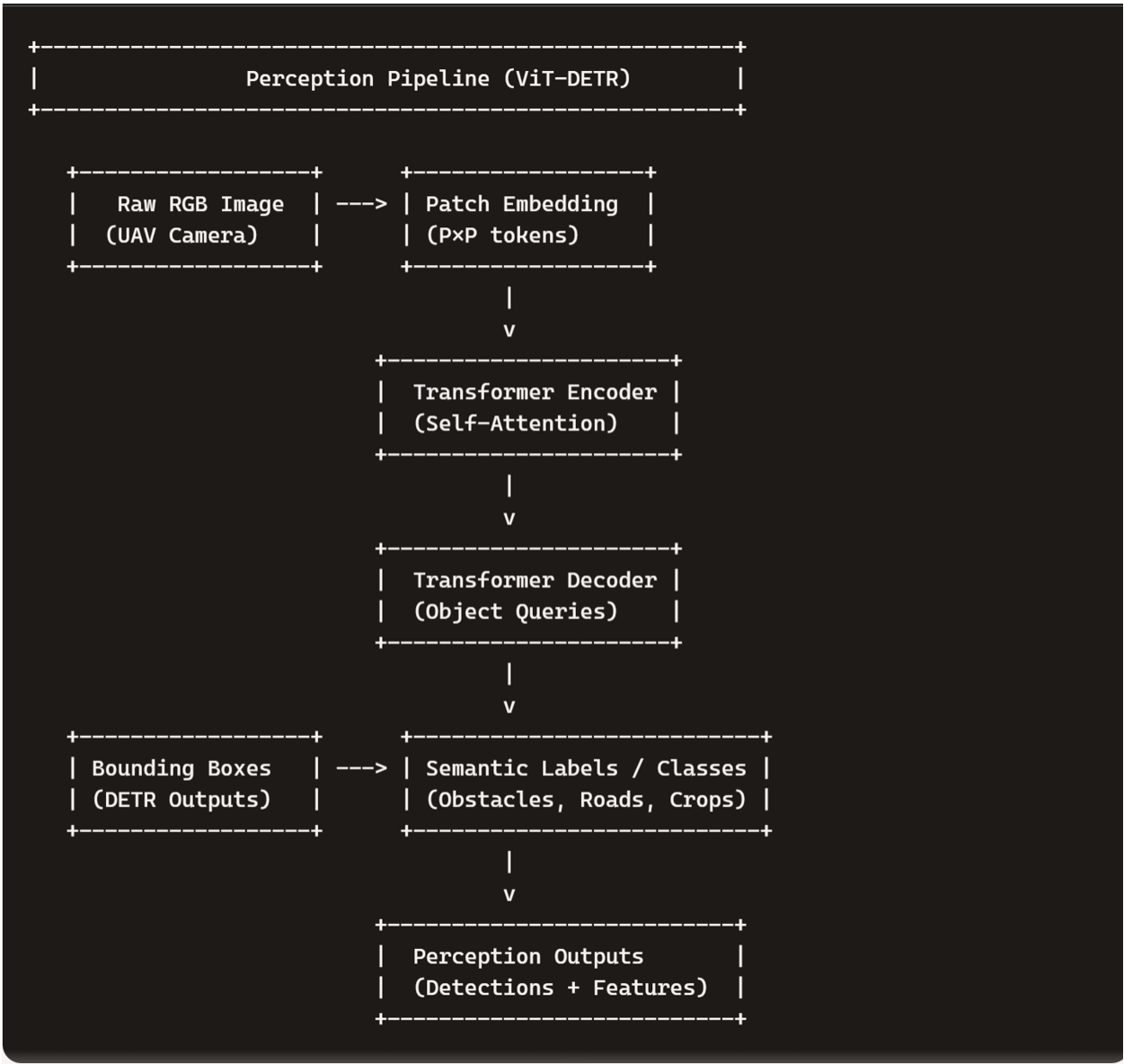


Fig. 2. ViT-DETR perception pipeline illustrating patch embedding, transformer encoding, decoding, and semantic output.

Factor-graph SLAM

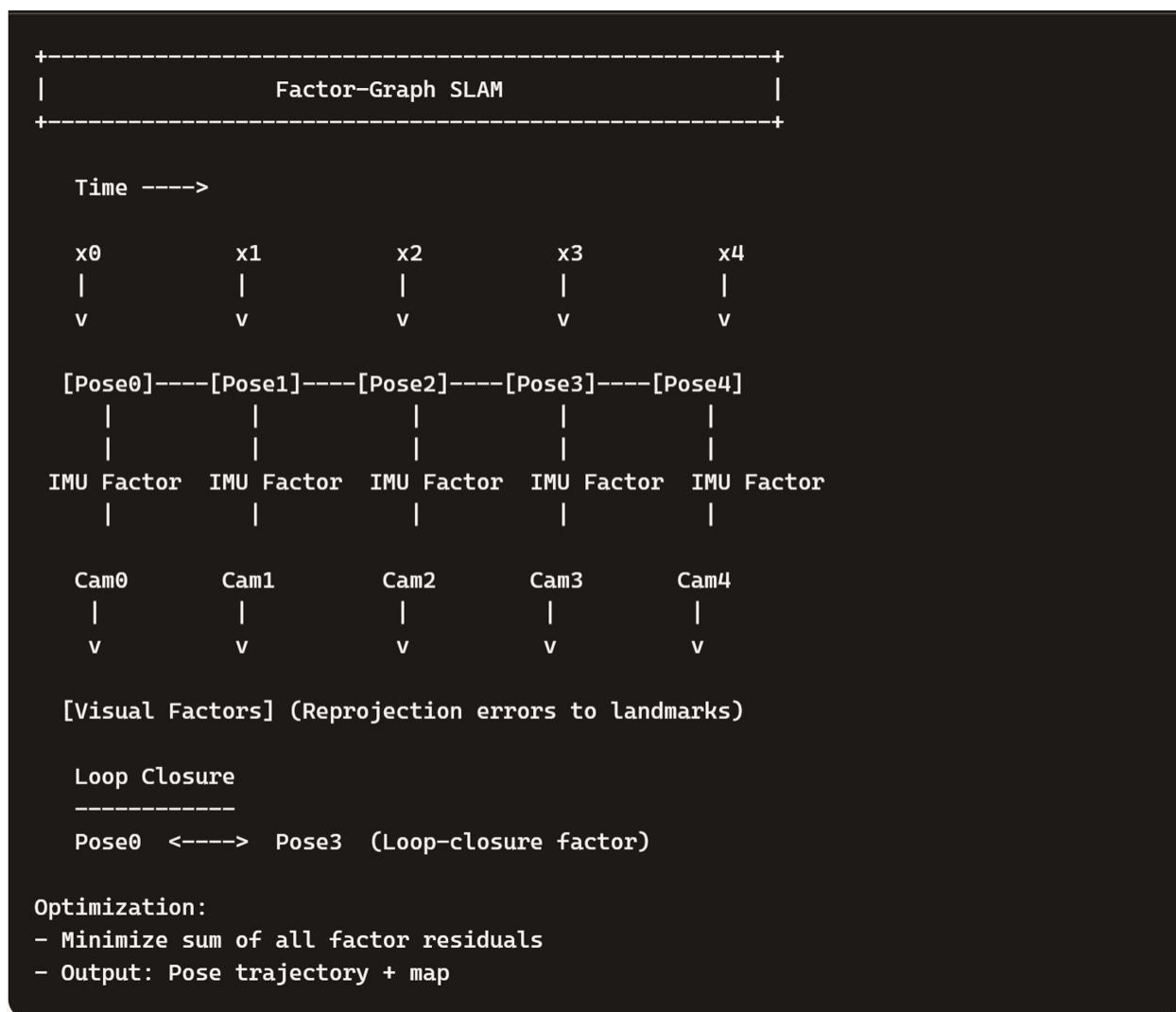


Fig.3. Factor-graph SLAM with IMU factors, visual factors, and loop-closure constraints.

MPC flow diagram



Fig.4. MPC flow diagram showing prediction, cost evaluation, constraint handling, and optimal control selection.

RL steering prior flowchart

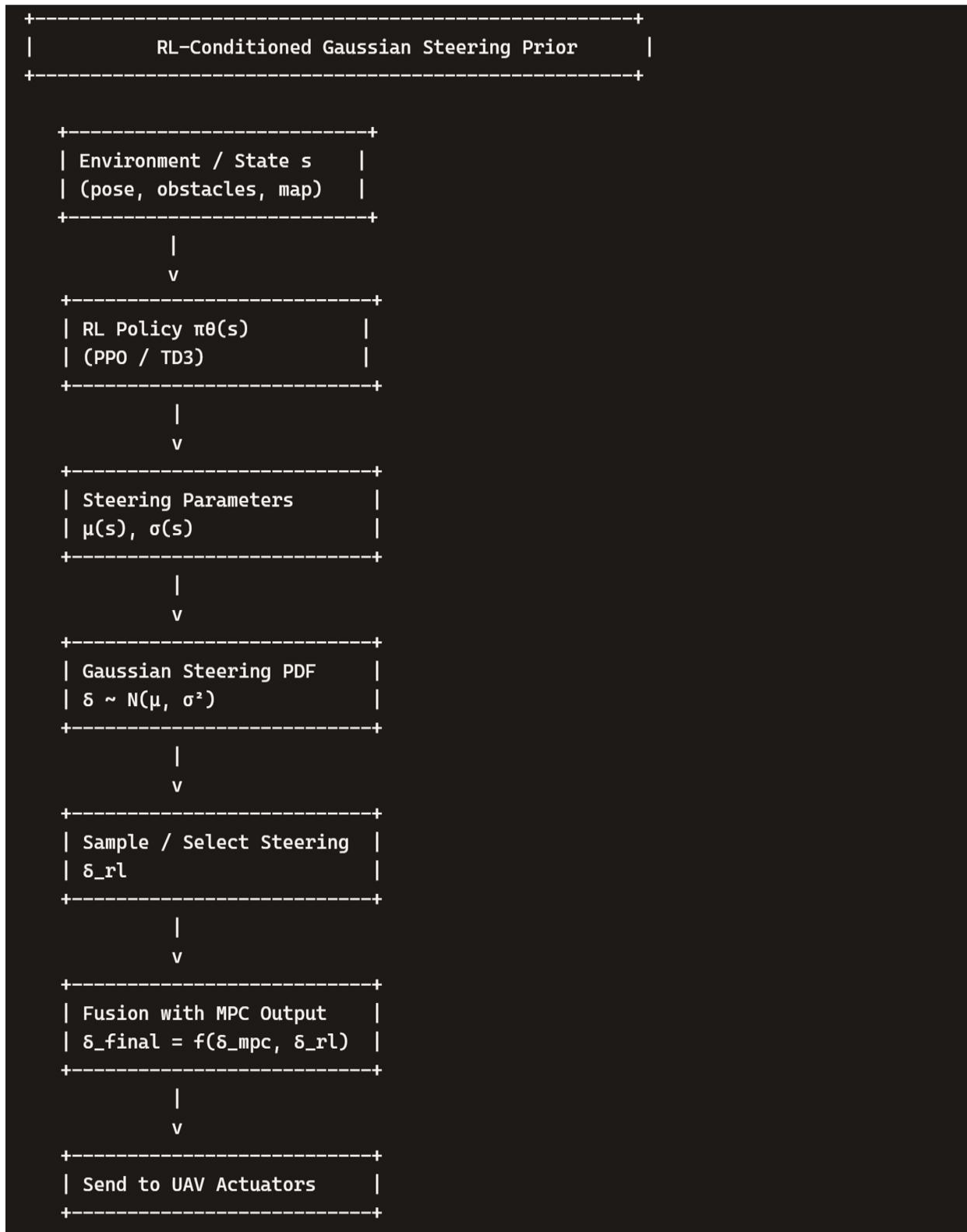


Fig.5. RL-conditioned Gaussian steering prior combining PPO/TD3 outputs with MPC steering.

Semantic mapping pipeline

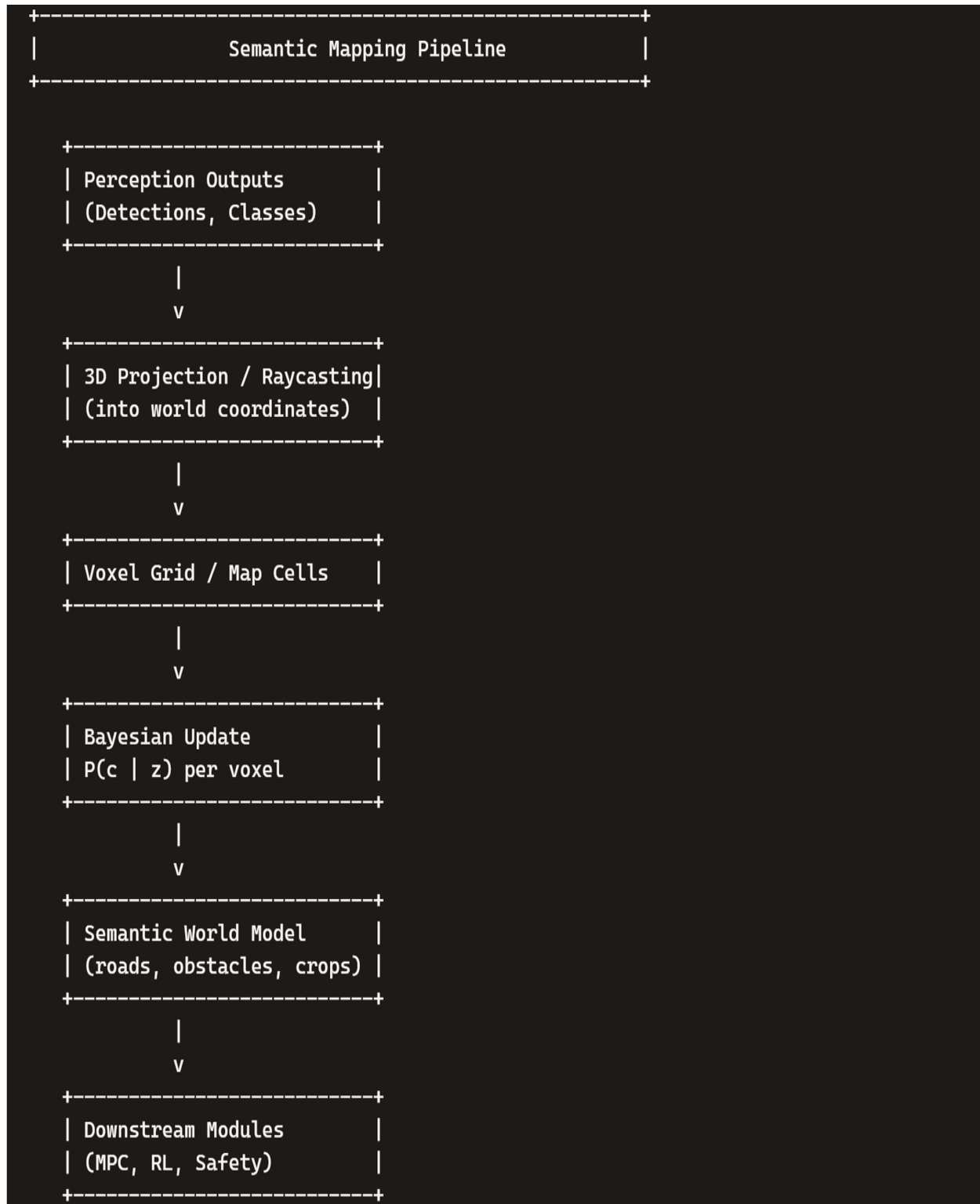


Fig.6. Semantic mapping pipeline using voxel grids and Bayesian updates to build a world model.

Full autonomy loop

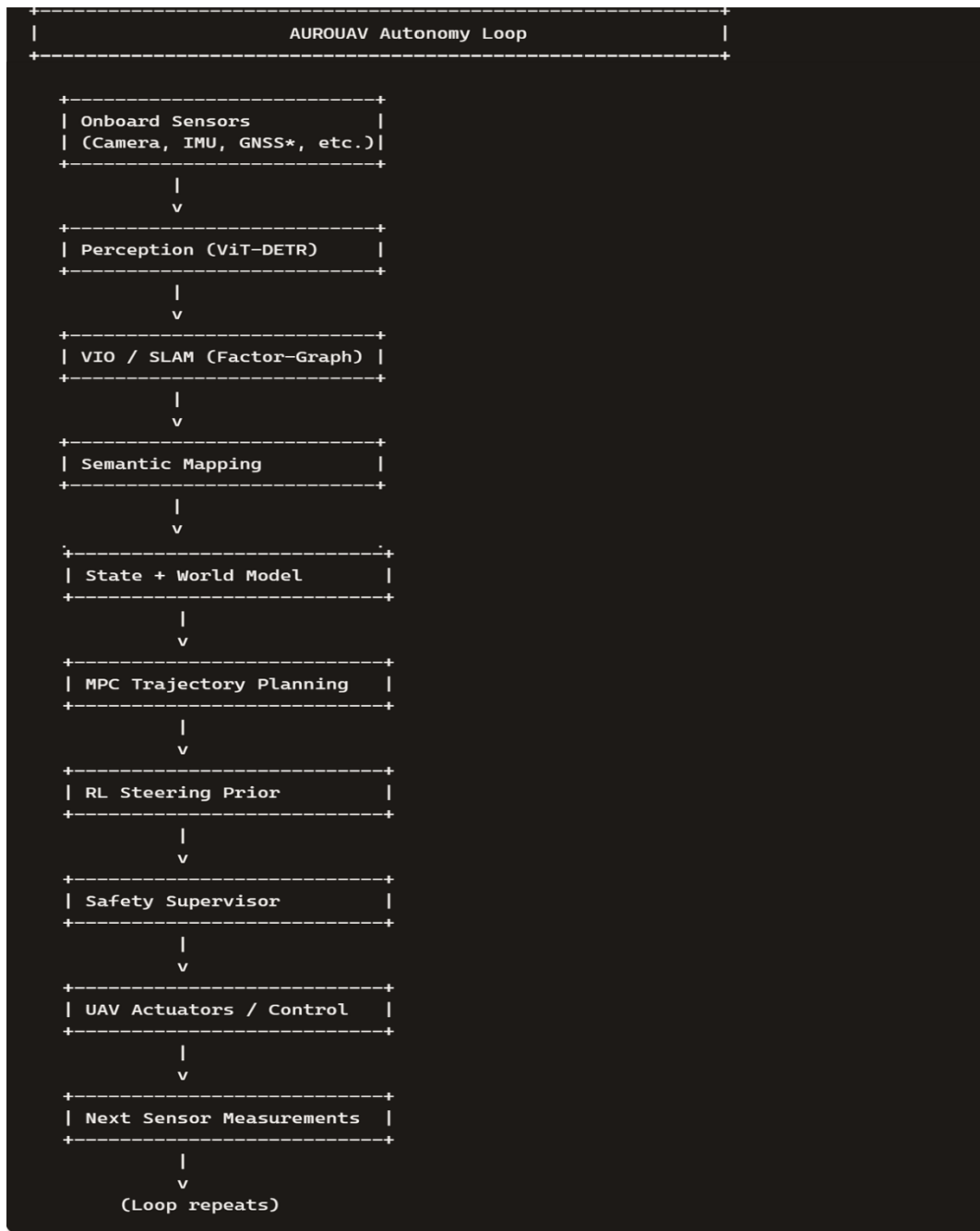


Fig.7. Full autonomy loop integrating sensing, perception, SLAM, mapping, planning, RL, safety, and actuation.